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JEE MAIN · QUESTION PAPER

JEE Main 2025 (Online) 4th April Evening Shift

JEE Main Question Paper with Answer Key

4 Apr 2025	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	25 questions
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Answer key	at the end

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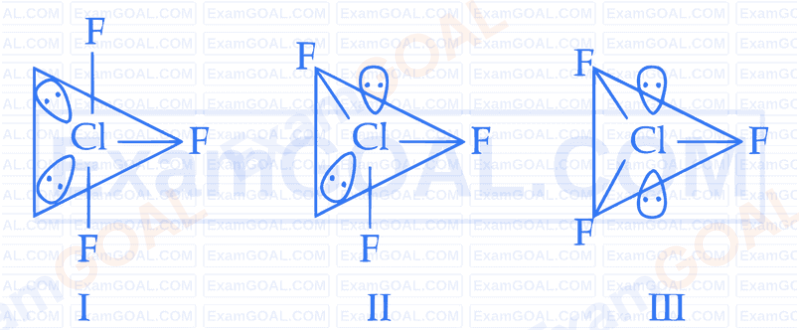
Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement (I): For  , all three possible structures may be drawn as follows.



Statement (II): Structure III is most stable, as the orbitals having the lone pairs are axial, where the lp – bp repulsion is minimum.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

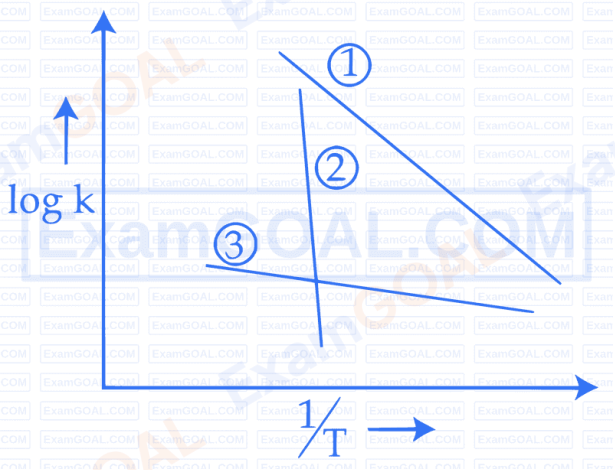
(D) Statement I is incorrect but Statement II is correct

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Q2. MCQ (Single Correct Answer)

Open solution

Consider the following plots of $\log$ of rate constant $k(\log k)$ vs $\frac{1}{T}$ for three different reactions. The correct order of activation energies of these reactions is :



(A) $Ea_2 > Ea_1 > Ea_3$

(B) $Ea_1 > Ea_3 > Ea_2$

(C) $Ea_3 > Ea_2 > Ea_1$

(D) $Ea_1 > Ea_2 > Ea_3$

Q3. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II

	List - I (Separation of)		**List - II (Separation Technique)**
(A)	Aniline from aniline-water mixture	(I)	Simple distillation
(B)	Glycerol from spent-lye in soap industry	(II)	Fractional distillation
(C)	Different fractions of crude oil in petroleum industry	(III)	Distillation at reduced pressure
(D)	Chloroform-Aniline mixture	(IV)	Steam distillation

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

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Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : The first ionisation enthalpy of group 14 elements is higher than the corresponding elements of group 13.

Statement (II) : Melting points and boiling points of group 13 elements are in general much higher than those of corresponding elements of group 14.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

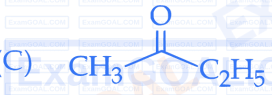
Q5. MCQ (Single Correct Answer)

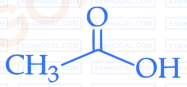
Open solution

Which among the following compounds give yellow solid when reacted with $\mathrm{NaOI/NaOH}$?

(A) 

(B) 

(C) 

(D) 

(E) 

Choose the correct answer from the options given below :

(A) (A) and (C) Only

(B) (C) and (D) Only

(C) (B), (C) and (E) Only

(D) (A), (C) and (D) Only

Q6. MCQ (Single Correct Answer)

Open solution

Consider the ground state of chromium atom $(Z = 24)$. How many electrons are with Azimuthal quantum number $l = 1$ and $l = 2$ respectively ?

(A) 16 and 5

(B) 12 and 5

(C) 12 and 4

(D) 16 and 4

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Q7. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement (I) : Molal depression constant K_f is given by $\frac{M_1RT_f}{\Delta S_{fus}}$, where symbols have their usual meaning.

Statement (II) : K_f for benzene is less than the K_f for water.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

Q8. MCQ (Single Correct Answer)

Open solution

The IUPAC name of the following compound is:



(A) 4-Hydroxyhept-6-en-1-yne

(B) Hept-1-en-6-yn-4-ol

(C) Hept-6-en-1-yn-4-ol

(D) 4-Hydroxyhept-1-en-6-yne

Q9. MCQ (Single Correct Answer)

Open solution

Half life of zero order reaction $A \rightarrow$ product is 1 hour, when initial concentration of reactant is $2.0\ mol\ L^{-1}$. The time required to decrease concentration of A from 0.50 to $0.25\ mol\ L^{-1}$ is :

(A) 0.5 hour

(B) 15 min

(C) 60 min

(D) 4 hour

Q10. MCQ (Single Correct Answer)

Open solution

The incorrect relationship in the following pairs in relation to ionisation enthalpies is :

(A) $Mn^{2+} < Fe^{2+}$

(B) $Mn^{+} < Mn^{2+}$

(C) $Mn^{+} < Cr^{+}$

(D) $Fe^{2+} < Fe^{3+}$

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Q11. MCQ (Single Correct Answer)

Open solution

A toxic compound " A " when reacted with NaCN in aqueous acidic medium yields an edible cooking component and food preservative " B ". " B " is converted to " C " by diborane and can be used as an additive to petrol to reduce emission. "C" upon reaction with oleum at $140^{\circ}C$ yields an inhalable anesthetic " D ". Identify " A ", " B ", " C " & " D ", respectively :

(A) Methanol; acetic acid; ethanol; diethyl ether

(B) Ethanol; acetonitrile; ethylamine; ethylene

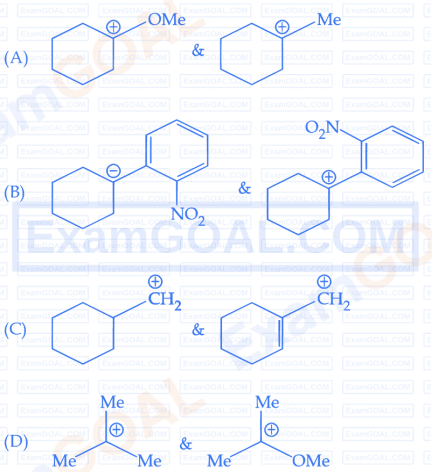
(C) Methanol; formaldehyde; methyl chloride; chloroform

(D) Acetaldehyde; 2-hydroxypropanoic acid; propanoic acid; dipropyl ether

Q12. MCQ (Single Correct Answer)

Open solution

In which pairs, the first ion is more stable than the second?



(A) (A) & (B) only **(B)** (B) & (D) only **(C)** (A) & (C) only **(D)** (B) & (C) only

Q13. MCQ (Single Correct Answer)

Open solution

' X ' is the number of electrons in t_{2g} orbitals of the most stable complex ion among $[Fe(NH_3)_6]^{3+}$, $[FeCl_6]^{3-}$, $[Fe(C_2O_4)_3]^{3-}$ and $[Fe(H_2O)_6]^{3+}$. The nature of oxide of vanadium of the type V_2O_X is :

(A) Amphoteric **(B)** Acidic **(C)** Basic **(D)** Neutral

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Q14. MCQ (Single Correct Answer)

Open solution

The correct order of basicity for the following molecules is :



(A) R > P > Q (B) P > Q > R (C) R > Q > P (D) Q > P > R

Q15. MCQ (Single Correct Answer)

Open solution

Consider the given data :

(a) $\mathrm{HCl(g) + 10H_2O(l) \rightarrow HCl.10H_2O \Delta H = -69.01\ kJ\ mol^{-1}}$

(b) $\mathrm{HCl(g) + 40H_2O(l) \rightarrow HCl.40H_2O \Delta H = -72.79\ kJ\ mol^{-1}}$

Choose the correct statement :

(A) The heat of dilution for the $\mathrm{HCl\ (HCl.10H_2O\ to\ HCl.40H_2O)}$ is $3.78\ \mathrm{kJ\ mol^{-1}}$.

(B) Dissolution of gas in water is an endothermic process.

(C) The heat of solution depends on the amount of solvent.

(D) The heat of formation of HCl solution is represented by both (a) and (b).

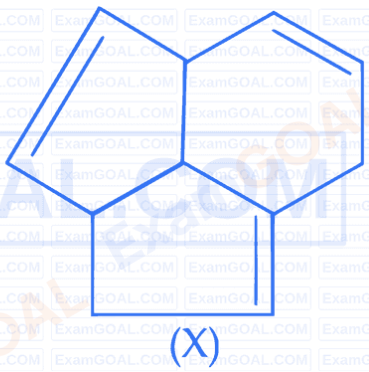
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Q16. MCQ (Single Correct Answer)

Open solution

Consider the following molecule(X).

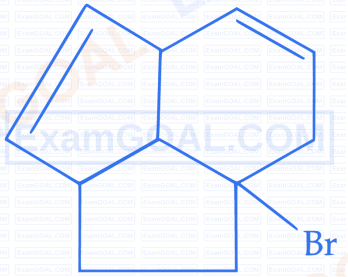
The structure of X is



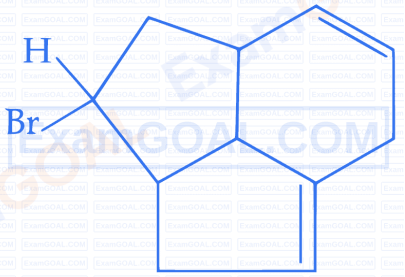
(X)

The major product formed when the given molecule (X) is treated with $\mathrm{HBr}(1\mathrm{eq})$ is :

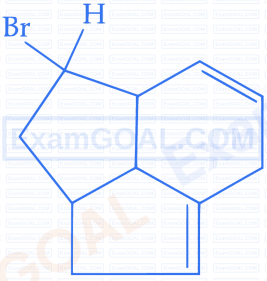
(A)



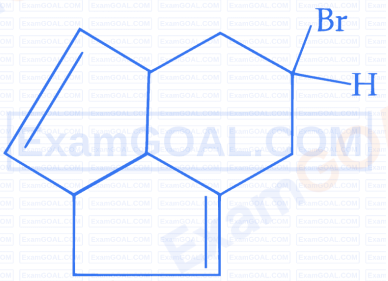
(B)



(C)



(D)



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Q17. MCQ (Single Correct Answer)

Open solution

The correct order of $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ complex species based on the number of unpaired electrons present is:

(A) $[CoF_6]^{3-} > [FeF_6]^{3-} > [Ni(CO)_4] > [Ni(CN)_4]^{2-}$

(B) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} > [Ni(CO)_4]$

(C) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} = [Ni(CO)_4]$

(D) $[Ni(CN)_4]^{2-} > [FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CO)_4]$

Q18. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : Alcohols are formed when alkyl chlorides are treated with aqueous potassium hydroxide by elimination reaction.

Statement (II) : In alcoholic potassium hydroxide, alkyl chlorides form alkenes by abstracting the hydrogen from the β-carbon.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are incorrect

(B) Both Statement I and Statement II are correct

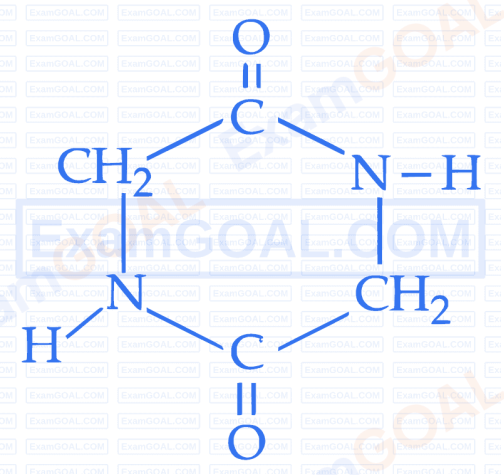
(C) Statement I is incorrect but Statement II is correct

(D) Statement I is correct but Statement II is incorrect

Q19. MCQ (Single Correct Answer)

Open solution

A dipeptide, " x " on complete hydrolysis gives " y " and " z ". " y " on treatment with aq. HNO_2 produces lactic acid. On the other hand " z " on heating gives the following cyclic molecule.



Based on the information given, the dipeptide X is :

(A) alanine-alanine (B) valine-glycine (C) valine-leucine (D) alanine-glycine

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Q20. MCQ (Single Correct Answer)

The elements of Group 13 with highest and lowest first ionisation enthalpies are respectively :

(A) B & Tl (B) Tl & B (C) B & In (D) B & Ga

Open solution

Q21. Numerical

Sea water, which can be considered as a 6 molar $(6\mathrm{M})$ solution of NaCl , has a density of $2\ \mathrm{g\ mL^{-1}}$. The concentration of dissolved oxygen $(\mathrm{O_2})$ in sea water is 5.8 ppm . Then the concentration of dissolved oxygen $(\mathrm{O_2})$ in sea water, in terms of molality, is $x \times 10^{-4}\ \mathrm{m}$.

$x =$ _____. (Nearest integer)

Given: Molar mass of NaCl is $58.5\ \mathrm{g\ mol^{-1}}$

Molar mass of $\mathrm{O_2}$ is $32\ \mathrm{g\ mol^{-1}}$

Open solution

Q22. Numerical

The amount of calcium oxide produced on heating 150 kg limestone (75% pure) is _____ kg. (Nearest integer)

Given: Molar mass (in $\mathrm{gmol^{-1}}$) of Ca-40, O-16, C-12

Open solution

Q23. Numerical

A metal complex with a formula $\mathrm{MCl_4 \cdot 3NH_3}$ is involved in $\mathrm{sp^3\ d^2}$ hybridisation. It upon reaction with excess of $\mathrm{AgNO_3}$ solution gives ' x ' moles of AgCl . Consider ' x ' is equal to the number of lone pairs of electron present in central atom of $\mathrm{BrF_5}$. Then the number of geometrical isomers exhibited by the complex is _____.

Open solution

Q24. Numerical

The molar conductance of an infinitely dilute solution of ammonium chloride was found to be $185\ \mathrm{S\ cm^2\ mol^{-1}}$ and the ionic conductance of hydroxyl and chloride ions are 170 and $70\ \mathrm{S\ cm^2\ mol^{-1}}$, respectively. If molar conductance of 0.02 M solution of ammonium hydroxide is $85.5\ \mathrm{S\ cm^2\ mol^{-1}}$, its degree of dissociation is given by $x \times 10^{-1}$. The value of x is _____ . (Nearest integer)

Open solution

Q25. Numerical

$x\mathrm{mg}$ of $\mathrm{Mg(OH)_2}$(molar mass $= 58$) is required to be dissolved in 1.0 L of water to produce a pH of 10.0 at 298 K . The value of x is _____ mg. (Nearest integer)

(Given : $\mathrm{Mg(OH)_2}$ is assumed to dissociate completely in $\mathrm{H_2O}$)

Open solution

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let $\mathrm{A}=\{-3,-2,-1,0,1,2,3\}$ and R be a relation on A defined by $x \mathrm{R} y$ if and only if $2x-y \in\{0,1\}$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l+\mathrm{m}+\mathrm{n}$ is equal to:

(A) 17 (B) 18 (C) 15 (D) 16

Open solution

Q27. MCQ (Single Correct Answer)

Let the domains of the functions $f(x)=\log_4 \log_3 \log_7\left(8-\log_2\left(x^2+4x+5\right)\right)$ and $\mathrm{g}(x)=\sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then $\alpha^2+\beta^2+\gamma^2+\delta^2$ is equal to :

(A) 15 (B) 13 (C) 16 (D) 14

Open solution

Q28. MCQ (Single Correct Answer)

Let the mean and the standard deviation of the observation $2,3,3,4,5,7, a, b$ be 4 and $\sqrt{2}$ respectively. Then the mean deviation about the mode of these observations is :

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) 1 (D) 2

Open solution

Q29. MCQ (Single Correct Answer)

Let the sum of the focal distances of the point $\mathrm{P}(4,3)$ on the hyperbola $\mathrm{H}: \frac{x^2}{\mathrm{a}^2}-\frac{y^2}{\mathrm{b}^2}=1$ be $8\sqrt{\frac{5}{3}}$. If for H , the length of the latus rectum is l and the product of the focal distances of the point P is m , then $9l^2+6\,\mathrm{m}$ is equal to :

(A) 187 (B) 184 (C) 186 (D) 185

Open solution

Q30. MCQ (Single Correct Answer)

Let $\mathrm{a}>0$. If the function $f(x)=6x^3-45\mathrm{a}x^2+108\mathrm{a}^2x+1$ attains its local maximum and minimum values at the points x_1 and x_2 respectively such that $x_1x_2=54$, then $\mathrm{a}+x_1+x_2$ is equal to :

(A) 15 (B) 13 (C) 24 (D) 18

Open solution

Q31. MCQ (Single Correct Answer)

Let the matrix $A=\begin{bmatrix}1 & 0 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$ satisfy $A^n=A^{n-2}+A^2-I$ for $n \geqslant 3$. Then the sum of all the elements of A^{50} is :

(A) 44 (B) 39 (C) 52 (D) 53

Open solution

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Q32. MCQ (Single Correct Answer)

If a curve $y = y(x)$ passes through the point $\left(1, \frac{\pi}{2}\right)$ and satisfies the differential equation $\left(7x^4 \cot y - e^x \operatorname{cosec} y\right) \frac{dx}{dy} = x^5, x \geq 1$, then at $x = 2$, the value of $\cos y$ is :

(A) $\frac{2e^2+e}{64}$ (B) $\frac{2e^2-e}{64}$

(C) $\frac{2e^2-e}{128}$ (D) $\frac{2e^2+e}{128}$

Open solution

Q33. MCQ (Single Correct Answer)

If the sum of the first 20 terms of the series $\frac{4 \cdot 1}{4+3 \cdot 1^2+1^4} + \frac{4 \cdot 2}{4+3 \cdot 2^2+2^4} + \frac{4 \cdot 3}{4+3 \cdot 3^2+3^4} + \frac{4 \cdot 4}{4+3 \cdot 4^2+4^4} + \ldots$ is $\frac{m}{n}$, where m and n are coprime, then $m + n$ is equal to :

(A) 423 (B) 421 (C) 422 (D) 420

Open solution

Q34. MCQ (Single Correct Answer)

Let for two distinct values of p the lines $y = x + p$ touch the ellipse $E : \frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$ at the points A and B . Let the line $y = x$ intersect E at the points C and D . Then the area of the quadrilateral $ABCD$ is equal to :

(A) 48 (B) 20 (C) 24 (D) 36

Open solution

Q35. MCQ (Single Correct Answer)

The centre of a circle C is at the centre of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$. Let C pass through the foci F_1 and F_2 of E such that the circle C and the ellipse E intersect at four points. Let P be one of these four points. If the area of the triangle PF_1F_2 is 30 and the length of the major axis of E is 17 , then the distance between the foci of E is :

(A) 12 (B) 26 (C) 13 (D) $\frac{13}{2}$

Open solution

Q36. MCQ (Single Correct Answer)

A line passing through the point $A(-2, 0)$, touches the parabola $P : y^2 = x - 2$ at the point B in the first quadrant. The area, of the region bounded by the line AB, parabola P and the x-axis, is :

(A) 3 (B) $\frac{7}{3}$ (C) $\frac{8}{3}$ (D) 2

Open solution

Q37. MCQ (Single Correct Answer)

Let the product of $\omega_1 = (8 + i) \sin\theta + (7 + 4i) \cos\theta$ and $\omega_2 = (1 + 8i) \sin\theta + (4 + 7i) \cos\theta$ be $\alpha + i\beta$, $i = \sqrt{-1}$. Let p and q be the maximum and the minimum values of $\alpha + \beta$ respectively. Then $p + q$ is equal to :

(A) 130 (B) 150 (C) 160 (D) 140

Open solution

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Q38. MCQ (Single Correct Answer)

If $1^2 \cdot \left({}^{15}C_1\right) + 2^2 \cdot \left({}^{15}C_2\right) + 3^2 \cdot \left({}^{15}C_3\right) + \ldots + 15^2 \cdot \left({}^{15}C_{15}\right) = 2^m \cdot 3^n \cdot 5^k$, where $m, n, k \in \mathbf{N}$, then $\mathrm{m} + \mathrm{n} + \mathrm{k}$ is equal to :

(A) 20 (B) 19 (C) 18 (D) 21

Open solution

Q39. MCQ (Single Correct Answer)

Let A be the point of intersection of the lines $\mathrm{L}_1 : \frac{x-7}{1} = \frac{y-5}{0} = \frac{z-3}{-1}$ and $\mathrm{L}_2 : \frac{x-1}{3} = \frac{y+3}{4} = \frac{z+7}{5}$. Let B and C be the points on the lines L_1 and L_2 respectively such that $AB = AC = \sqrt{15}$. Then the square of the area of the triangle ABC is :

(A) 63 (B) 57 (C) 60 (D) 54

Open solution

Q40. MCQ (Single Correct Answer)

The sum of the infinite series $\cot^{-1}\left(\frac{7}{4}\right) + \cot^{-1}\left(\frac{19}{4}\right) + \cot^{-1}\left(\frac{39}{4}\right) + \cot^{-1}\left(\frac{67}{4}\right) + \ldots.$ is :

(A) $\frac{\pi}{2} + \cot^{-1}\left(\frac{1}{2}\right)$ (B) $\frac{\pi}{2} - \cot^{-1}\left(\frac{1}{2}\right)$

(C) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{2}\right)$ (D) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{2}\right)$

Open solution

Q41. MCQ (Single Correct Answer)

The axis of a parabola is the line $y = x$ and its vertex and focus are in the first quadrant at distances $\sqrt{2}$ and $2\sqrt{2}$ units from the origin, respectively. If the point $(1, k)$ lies on the parabola, then a possible value of k is :

(A) 8 (B) 3 (C) 9 (D) 4

Open solution

Q42. MCQ (Single Correct Answer)

Let $f(x) + 2f\left(\frac{1}{x}\right) = x^2 + 5$ and $2g(x) - 3g\left(\frac{1}{2}\right) = x, x > 0$. If $\alpha = \int_1^2 f(x)\mathrm{d}x$, and $\beta = \int_1^2 g(x)\mathrm{d}x$, then the value of $9\alpha + \beta$ is :

(A) 0 (B) 10 (C) 1 (D) 11

Open solution

Q43. MCQ (Single Correct Answer)

Consider two sets A and B, each containing three numbers in A.P. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and q respectively. Let d and D be the common differences of $\mathrm{AP}'\mathrm{s}$ in A and B respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$, then $\mathrm{p} - \mathrm{q}$ is equal to

(A) 540 (B) 450 (C) 600 (D) 630

Open solution

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Q44. MCQ (Single Correct Answer)

Let the values of p , for which the shortest distance between the lines $\frac{x+1}{3} = \frac{y}{4} = \frac{z}{5}$ and $\vec{r} = (p\hat{i} + 2\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ is $\frac{1}{\sqrt{6}}$, be $a, b, (a < b)$. Then the length of the latus rectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :

(A) $\frac{3}{2}$ (B) 9 (C) 18 (D) $\frac{2}{3}$

Open solution

Q45. MCQ (Single Correct Answer)

Let f be a differentiable function on $\mathbf{R}$ such that $f(2) = 1, f'(2) = 4$. Let $\lim_{x\to 0}(f(2+x))^{3/x} = e^{\alpha}$. Then the number of times the curve $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$ meets x-axis is :

(A) 3 (B) 1 (C) 2 (D) 0

Open solution

Q46. Numerical

If $\int \frac{\left(\sqrt{1+x^2}+x\right)^{10}}{\left(\sqrt{1+x^2}-x\right)^{9}} dx = \frac{1}{m}\left(\left(\sqrt{1+x^2}+x\right)^{n}\left(n\sqrt{1+x^2}-x\right)\right) + C$ where C is the constant of integration and $m, n \in \mathbf{N}$, then $m + n$ is equal to _____ .

Open solution

Q47. Numerical

A card from a pack of 52 cards is lost. From the remaining 51 cards, n cards are drawn and are found to be spades. If the probability of the lost card to be a spade is $\frac{11}{50}$, then n is equal to _____ .

Open solution

Q48. Numerical

Let the three sides of a triangle ABC be given by the vectors $2\hat{i} - \hat{j} + \hat{k}, \hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$. Let G be the centroid of the triangle ABC. Then $6\left(|\overrightarrow{AG}|^2 + |\overrightarrow{BG}|^2 + |\overrightarrow{CG}|^2\right)$ is equal to _____.

Open solution

Q49. Numerical

If α is a root of the equation $x^2 + x + 1 = 0$ and $\sum_{k=1}^{n}\left(\alpha^k + \frac{1}{\alpha^k}\right)^2 = 20$, then n is equal to _____.

Open solution

Q50. Numerical

Let m and $n, (m < n)$, be two 2-digit numbers. Then the total numbers of pairs (m, n), such that $\gcd(m, n) = 6$, is _____ .

Open solution

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

A finite size object is placed normal to the principal axis at a distance of 30 cm from a convex mirror of focal length 30 cm . A plane mirror is now placed in such a way that the image produced by both the mirrors coincide with each other. The distance between the two mirrors is :

Open solution

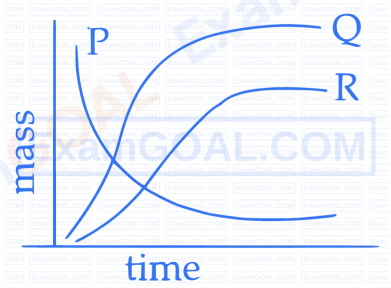
(A) 45 cm (B) 15 cm (C) 22.5 cm (D) 7.5 cm

Q52. MCQ (Single Correct Answer)

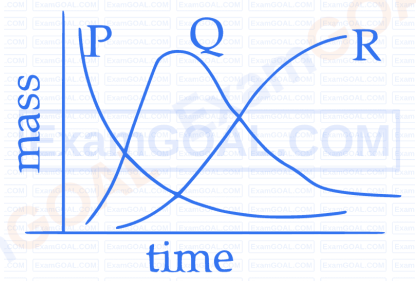
A radioactive material P first decays into Q and then Q decays to non-radioactive material R. Which of the following figure represents time dependent mass of P, Q and R ?

Open solution

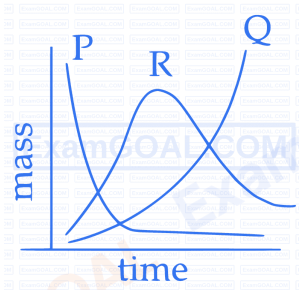
(A)



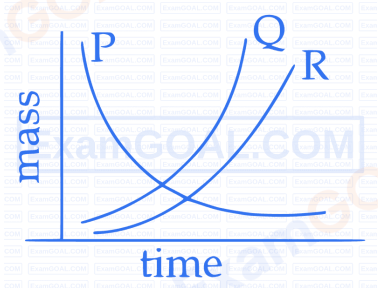
(B)



(C)



(D)



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Q53. MCQ (Single Correct Answer)

Displacement of a wave is expressed as $x(t) = 5\cos\left(628t + \frac{\pi}{2}\right)\text{m}$. The wavelength of the wave when its velocity is $300\ \text{m/s}$ is :

$(\pi = 3.14)$

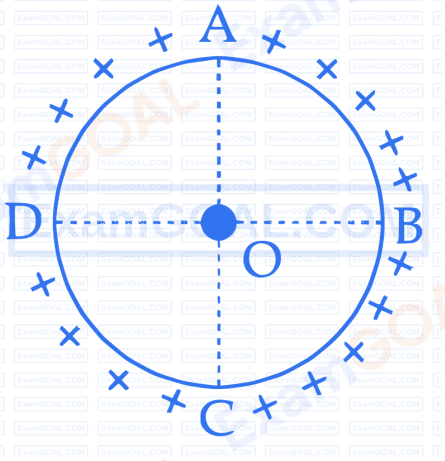
Open solution

(A) 0.33 m (B) 0.5 m (C) 3 m (D) 5 m

Q54. MCQ (Single Correct Answer)

A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc AB at ' O ' is ' E ' in magnitude. What would be the magnitude of electric field at ' O ' due to arc ABC ?

Open solution



(A) 2E (B) Zero (C) E/2 (D) $\sqrt{2}$E

Q55. MCQ (Single Correct Answer)

A wheel is rolling on a plane surface. The speed of a particle on the highest point of the rim is $8\ \text{m/s}$. The speed of the particle on the rim of the wheel at the same level as the centre of wheel, will be :

Open solution

(A) $4\sqrt{2}\ \text{m/s}$ (B) $8\ \text{m/s}$

(C) $4\ \text{m/s}$ (D) $8\sqrt{2}\ \text{m/s}$

Q56. MCQ (Single Correct Answer)

In an electromagnetic system, a quantity defined as the ratio of electric dipole moment and magnetic dipole moment has dimension of $\left[\text{M}^{\text{P}}\text{L}^{\text{Q}}\text{T}^{R}A^{\text{S}}\right]$. The value of P and Q are :

Open solution

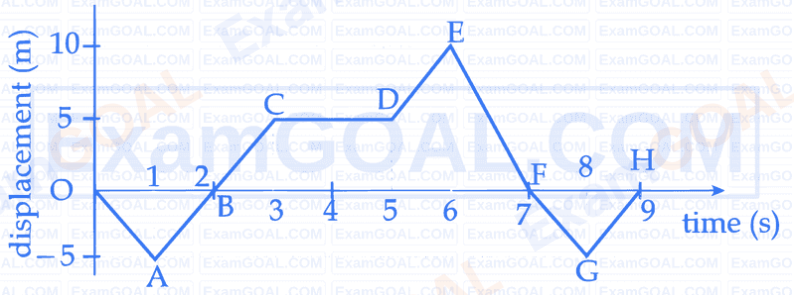
(A) $-1, 0$ (B) $0, -1$ (C) $-1, 1$ (D) $1, -1$

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Q57. MCQ (Single Correct Answer)

Open solution

The displacement x versus time graph is shown below.



(A) The average velocity during 0 to 3 s is $10\ \mathrm{m/s}$

(B) The average velocity during 3 to 5 s is $0\ \mathrm{m/s}$

(C) The instantaneous velocity at $\mathrm{t} = 2\ \mathrm{s}$ is $5\ \mathrm{m/s}$

(D) The average velocity during 5 to 7 s and instantaneous velocity at $\mathrm{t} = 6.5\ \mathrm{s}$ are equal

(E) The average velocity from $t = 0$ to $t = 9\ \mathrm{s}$ is zero

Choose the correct answer from the options given below :

(A) (B), (C), (D) only **(B)** (B), (D), (E) only

(C) (A), (D), (E) only **(D)** (B), (C), (E) only

Q58. MCQ (Single Correct Answer)

Open solution

Consider a rectangular sheet of solid material of length $l = 9\ \mathrm{cm}$ and width $\mathrm{d} = 4\ \mathrm{cm}$. The coefficient of linear expansion is $\alpha = 3.1 \times 10^{-5}\ \mathrm{K}^{-1}$ at room temperature and one atmospheric pressure. The mass of sheet $m = 0.1\ \mathrm{kg}$ and the specific heat capacity $C_{\mathrm{v}} = 900\ \mathrm{J\ kg^{-1}\ K^{-1}}$. If the amount of heat supplied to the material is $8.1 \times 10^2\ \mathrm{J}$ then change in area of the rectangular sheet is :

(A) $2.0 \times 10^{-6}\ \mathrm{m}^2$ **(B)** $6.0 \times 10^{-7}\ \mathrm{m}^2$ **(C)** $3.0 \times 10^{-7}\ \mathrm{m}^2$ **(D)** $4.0 \times 10^{-7}\ \mathrm{m}^2$

Q59. MCQ (Single Correct Answer)

Open solution

An object is kept at rest at a distance of $3R$ above the earth's surface where R is earth's radius. The minimum speed with which it must be projected so that it does not return to earth is : (Assume $\mathrm{M} =$ mass of earth, $\mathrm{G} =$ Universal gravitational constant)

(A) $\sqrt{\frac{3\mathrm{GM}}{\mathrm{R}}}$ **(B)** $\sqrt{\frac{2\mathrm{GM}}{\mathrm{R}}}$

(C) $\sqrt{\frac{\mathrm{GM}}{2\mathrm{R}}}$ **(D)** $\sqrt{\frac{\mathrm{GM}}{\mathrm{R}}}$

Q60. MCQ (Single Correct Answer)

Open solution

A cylindrical rod of length 1 m and radius 4 cm is mounted vertically. It is subjected to a shear force of $10^5\ \mathrm{N}$ at the top. Considering infinitesimally small displacement in the upper edge, the angular displacement θ of the rod axis from its original position would be : (shear moduli, $G = 10^{10}\ \mathrm{N/m^2}$)

(A) $1/160\pi$ **(B)** $1/2\pi$ **(C)** $1/4\pi$ **(D)** $1/40\pi$

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Q61. MCQ (Single Correct Answer)

For the determination of refractive index of glass slab, a travelling microscope is used whose main scale contains 300 equal divisions equals to 15 cm . The vernier scale attached to the microscope has 25 divisions equals to 24 divisions of main scale. The least count (LC) of the travelling microscope is (in cm) :

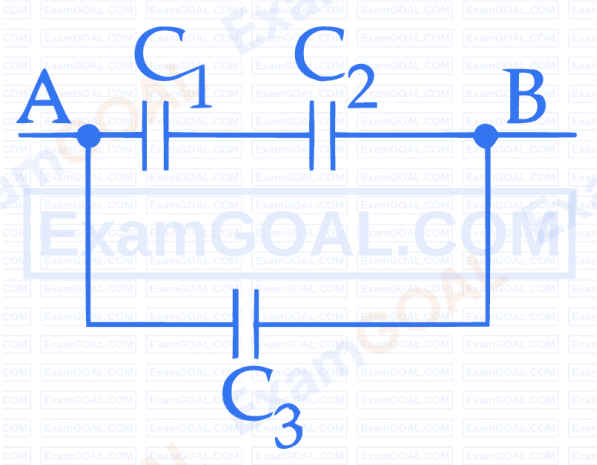
Open solution

(A) 0.002 (B) 0.0025 (C) 0.0005 (D) 0.001

Q62. MCQ (Single Correct Answer)

Three parallel plate capacitors C_1, C_2 and C_3 each of capacitance $5\mu\,\mathrm{F}$ are connected as shown in figure. The effective capacitance between points A and B, when the space between the parallel plates of C_1 capacitor is filled with a dielectric medium having dielectric constant of 4, is :

Open solution



(A) $9\mu\,\mathrm{F}$ (B) $30\mu\,\mathrm{F}$ (C) $7.5\mu\,\mathrm{F}$ (D) $22.5\mu\,\mathrm{F}$

Q63. MCQ (Single Correct Answer)

A block of mass 25 kg is pulled along a horizontal surface by a force at an angle 45° with the horizontal. The friction coefficient between the block and the surface is 0.25 . The block travels at a uniform velocity. The work done by the applied force during a displacement of 5 m of the block is :

Open solution

(A) 490 J (B) 970 J (C) 735 J (D) 245 J

Q64. MCQ (Single Correct Answer)

There are two vessels filled with an ideal gas where volume of one is double the volume of other. The large vessel contains the gas at 8 kPa at 1000 K while the smaller vessel contains the gas at 7 kPa at 500 K . If the vessels are connected to each other by a thin tube allowing the gas to flow and the temperature of both vessels is maintained at 600 K , at steady state the pressure in the vessels will be (in kPa).

Open solution

(A) 24 (B) 4.4 (C) 18 (D) 6

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Q65. MCQ (Single Correct Answer)

Open solution

Two polarisers P_1 and P_2 are placed in such a way that the intensity of the transmitted light will be zero. A third polariser P_3 is inserted in between P_1 and P_2, at particular angle between P_2 and P_3. The transmitted intensity of the light passing through all three polarisers is maximum. The angle between the polarisers P_2 and P_3 is :

(A) $\pi/6$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\pi/8$

Q66. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I		List - II
(A)	Isobaric	(I)	$\Delta Q = \Delta W$
(B)	Isochoric	(II)	$\Delta Q = \Delta U$
(C)	Adiabatic	(III)	$\Delta Q =$ zero
(D)	Isothermal	(IV)	$\Delta Q = \Delta U + P\Delta V$

$\Delta Q =$ Heat supplied

$\Delta W =$ Work done by the system

$\Delta \mathrm{U} =$ Change in internal energy

$\mathrm{P} =$ Pressure of the system

$\Delta \mathrm{V} =$ Change in volume of the system

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(B) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(D) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Q67. MCQ (Single Correct Answer)

Open solution

There are ' n ' number of identical electric bulbs, each is designed to draw a power p independently from the mains supply. They are now joined in series across the mains supply. The total power drawn by the combination is :

(A) np (B) p (C) $\frac{\mathrm{p}}{\mathrm{n}^2}$ (D) $\frac{p}{n}$

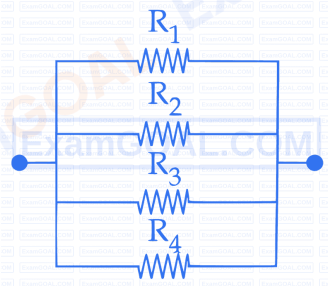
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Q68. MCQ (Single Correct Answer)

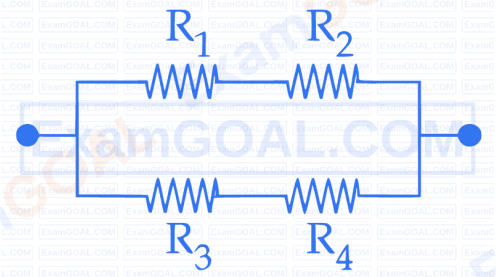
Open solution

From the combination of resistors with resistances values $R_1 = R_2 = R_3 = 5\Omega$ and $R_4 = 10\Omega$, which of the following combination is the best circuit to get an equivalent resistance of 6Ω ?

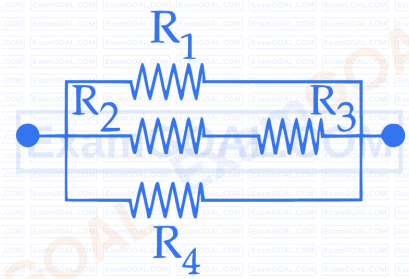
(A)



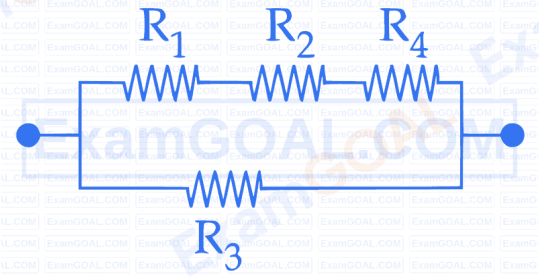
(B)



(C)



(D)



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Q69. MCQ (Single Correct Answer)

Open solution

Consider a n-type semiconductor in which n_e and n_h are number of electrons and holes, respectively.

(A) Holes are minority carriers

(B) The dopant is a pentavalent atom

(C) $n_e n_h \neq n_i^2$

(where n_i is number of electrons or holes in semiconductor when it is intrinsic form)

(D) $n_e n_h \geqslant n_i^2$

(E) The holes are not generated due to the donors

Choose the correct answer from the options given below :

(A) (A), (B), (E) only

(B) (A), (C), (E) only

(C) (A), (B), (C) only

(D) (A), (C), (D) only

Q70. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : The dimensions of Planck's constant and angular momentum are same.

Statement (II) : In Bohr's model electron revolve around the nucleus only in those orbits for which angular momentum is integral multiple of Planck's constant.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

Q71. Numerical

Open solution

A solid sphere with uniform density and radius R is rotating initially with constant angular velocity (ω_1) about its diameter. After some time during the rotation it starts losing mass at a uniform rate, with no change in its shape. The angular velocity of the sphere when its radius become $R/2$ is $x\omega_1$. The value of x is _____.

Q72. Numerical

Open solution

An inductor of self inductance 1 H is connected in series with a resistor of 100π ohm and an ac supply of 100π volt, 50 Hz . Maximum current flowing in the circuit is _____ A.

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Q73. Numerical

If an optical medium possesses a relative permeability of $\frac{10}{\pi}$ and relative permittivity of $\frac{1}{0.0885}$, then the velocity of light is greater in vacuum than that in this medium by _____ times.

$\left(\mu_0 = 4\pi \times 10^{-7} \mathrm{H/m}, \epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F/m}, \mathrm{c} = 3 \times 10^8\ \mathrm{m/s}\right)$

Open solution

Q74. Numerical

A particle of charge $1.6\mu\mathrm{C}$ and mass $16\mu\ \mathrm{g}$ is present in a strong magnetic field of 6.28 T . The particle is then fired perpendicular to magnetic field. The time required for the particle to return to original location for the first time is _____ s. $(\pi = 3.14)$

Open solution

Q75. Numerical

In a Young's double slit experiment, two slits are located 1.5 mm apart. The distance of screen from slits is 2 m and the wavelength of the source is 400 nm . If the 20 maxima of the double slit pattern are contained within the central maximum of the single slit diffraction pattern, then the width of each slit is $x \times 10^{-3}\ \mathrm{cm}$, where x-value is _____ .

Open solution

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Answer Key

Chemistry

1. **C**	2. **A**	3. **A**	4. **A**	5. **A**
6. **B**	7. **A**	8. **B**	9. **B**	10. **A**
11. **A**	12. **A**	13. **A**	14. **C**	15. **C**
16. **A**	17. **C**	18. **C**	19. **D**	20. **C**
21. **2**	22. **63**	23. **2**	24. **3**	25. **3**

Mathematics

26. **A**	27. **A**	28. **C**	29. **D**	30. **D**
31. **D**	32. **C**	33. **B**	34. **C**	35. **C**
36. **C**	37. **A**	38. **B**	39. **D**	40. **C**
41. **C**	42. **D**	43. **A**	44. **D**	45. **C**
46. **379**	47. **2**	48. **164**	49. **11**	50. **64**

Physics

51. **D**	52. **B**	53. **C**	54. **D**	55. **A**
56. **B**	57. **D**	58. **A**	59. **C**	60. **A**
61. **A**	62. **A**	63. **D**	64. **D**	65. **C**
66. **B**	67. **D**	68. **B**	69. **A**	70. **B**
71. **32**	72. **1**	73. **6**	74. **0**	75. **15**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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